Surface-polaritonlike waves guided by thin, lossy metal films

G. I. Stegeman and J. J. Burke

Optical Sciences Center, University of Arizona, Tucson, Arizona 85721

D. G. Hall

Institute of Optics, University of Rochester, Rochester, New York 14627

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Surface-plasmon polaritons guided by thin, lossy metallic films bounded by dissimilar dielectric media are investigated. New solutions to the dispersion relation are found, representing waves that are leaky (radiative) in one of the dielectrics. The new waves are interpreted in terms of the coupling of a damped surface plasmon at one interface with continuum modes at the other. Their excitation by end-fire coupling techniques is suggested.

Surface polaritons, such as surface plasmons, are electromagnetic waves that are guided by the interfaces between two or more media, one of which (the polariton-active medium) is characterized by a negative complex dielectric constant.[1] For many years[1,2] it was widely believed that the classic case of a thin metal film supports two surface-plasmon modes. Recently[3] Ferguson *et al.* showed that the lower branch splits into two in a Kretschmann geometry. In this Letter we show that a thin, lossy metal film bounded by dissimilar dielectric media can always support a total of four waves. Leaky waves are predicted in regions of the dispersion curves usually associated only with bound modes. (Although we discuss in detail surface-plasmon polaritons, the results and trends are also valid for other thin-film surface-polariton modes characterized by a negative complex dielectric constant, such as surface exciton polaritons[4] and surface phonon polaritons.[5]) Much of the previous surface-plasmon work[6–8] was focused on plasmons launched and scattered by surface roughness or by attenuated-total-reflection (ATR) arrangements; the new leaky waves, on the other hand, should be more readily observable under end-fire launching conditions.

The geometry that we consider consists of a thin metal film of dielectric constant $\epsilon_m = -\epsilon_R - i\epsilon_I$ bounded at $z = 0$ and $z = h$, respectively, by media with real dielectric constants ϵ_1 and ϵ_3 ($\epsilon_1 > \epsilon_3$). Waves supported by this geometry are described by $f(z)\exp(i\omega t - i\beta x)$ and are solutions to the dispersion relation $\tan(s_2h) = [s_2\epsilon_m(s_1\epsilon_3 + s_3\epsilon_1)]/(s_2{}^2\epsilon_1\epsilon_3 - s_1s_3\epsilon_m{}^2)$, where $s_1{}^2 = \beta^2 - \epsilon_1k^2$, $s_2{}^2 = \epsilon_mk^2 - \beta^2$, and $s_3{}^2 = \beta_2 - \epsilon_3k^2$. Here $f(z)$ describes the transverse field distribution and $\beta = \beta_R - \beta_I$ is the complex propagation wave vector. When $h \to \infty$, there are two decoupled surface-plasmon polariton modes localized at the ϵ_1–ϵ_m and ϵ_3–ϵ_m interfaces with propagation wave vectors $\beta_1(\infty)$ and $\beta_3(\infty)$, respectively.

The nature of the thin-film solutions can be classified[9] in terms of the fields in the dielectric media, $\exp(s_1z)$ and $\exp[-s_3(z - h)]$, respectively, with $s_1 = s_{1R} - is_{1I}$ and $s_3 = s_{3R} - is_{3I}$. For $S_{1R}, s_{1I}, s_{3R}, s_{3I} > 0$, the fields decay exponentially perpendicularly to the metal, as is characteristic of bound modes. As is indicated in Fig. 1, the planes of constant phase in the dielectrics (which are perpendicular to the planes of constant amplitude) are tilted toward the film where energy is dissipated by virtue of $\epsilon_I \neq 0$. A leaky-mode solution occurs for $s_{1R,I} < 0$ and $s_{3R,I} > 0$ (or $s_{1R,I} > 0$, $s_{3R,I} < 0$) and corresponds to a wave localized at the ϵ_3–ϵ_m ($\epsilon_1 - \epsilon_m$ for the second case) interface that couples to, and loses power, to outgoing radiation fields in medium $\epsilon_1(\epsilon_3)$. The conditions $s_{1R} < 0$ and $s_{1I}, s_{3R,I} > 0$ correspond to a wave localized at the ϵ_3–ϵ_m interface, which grows with propagation distance ($\beta_I < 0$), i.e., it is driven by an appropriately contoured radiation field incident from medium ϵ_1. No other solutions yield a physically rational interpretation.

For a symmetrically bounded ($\epsilon_1 = \epsilon_3$) metal film, only two bound Fano modes occur; no leaky waves can exist that radiate to one side only. This is a consequence of symmetry for this geometry, which requires

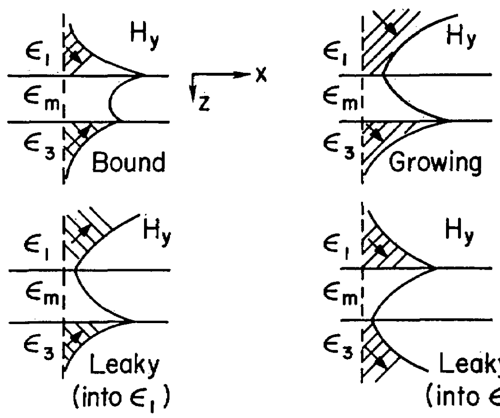

Fig. 1. Field distributions associated with bound, leaky, and growing modes guided by thin metal films. The arrows indicate energy flow in the dielectrics.

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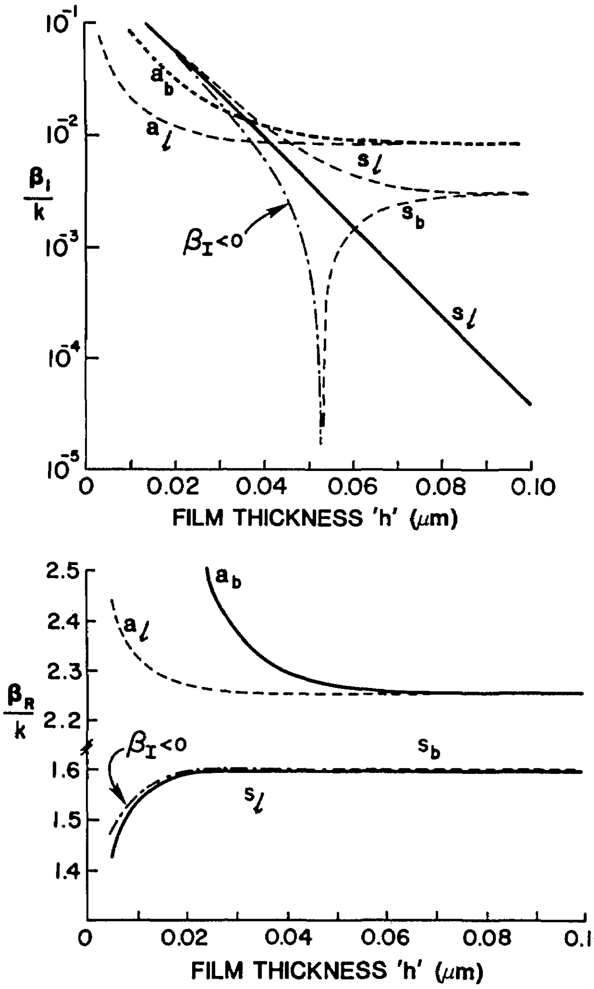

Fig. 2. The dispersion curves for a metal film of thickness h with $\epsilon_1 = 4.0$ and $\epsilon_3 = 2.25$, $\lambda = 0.633$ μm, and $\epsilon_m = -19 - 0.53i$; s and a refer to symmetric and antisymmetric modes, respectively. The solid lines are for $\epsilon_I = 0$ as well as $\epsilon_I \neq 0$, and the dashed lines show the new features that appear when $\epsilon_I \neq 0$. The subscripts b and l refer to bound and leaky modes, respectively.

that the field distribution exhibit symmetry with respect to the center of the film. Consequently the bound Fano modes have either a symmetric $[(\sqrt{\epsilon_1}k < \beta_R < \beta_{1R}(\infty)]$ or an antisymmetric $[(\beta_R > \beta_{1R}(\infty)$; zero field at $z = h/2)]$ transverse field (E_z).

For asymmetrically bounded $(\epsilon_1 > \epsilon_3)$ metal films, bound, leaky, and growing $(\beta_I < 0)$ modes are in principle allowed by lack of symmetry. It is widely believed[1] that there are still two Fano modes, one antisymmetric (i.e., $E_z = 0$ for some z in $0 \leq z \leq h$) and one symmetric ($E_z \neq 0$ for $0 \leq z \leq h$). The antisymmetric branch (as in Figs. 2 and 3) is bound for all values of ϵ_1 and ϵ_3. For $\beta_3(\infty) > \sqrt{\epsilon_1}k$, the symmetric mode is bound for a range of thickness $h > h_{co}$ (Fig. 3), but for $\sqrt{\epsilon_1}k > \beta_3(\infty)$ this Fano mode becomes leaky (s_l, Fig. 2) for all film thicknesses.

We find that this picture is incomplete and in some aspects incorrect. There are in all four physically significant solutions to the dispersions relations as long as $\epsilon_1 \neq \epsilon_3$. Ferguson *et al.* recently showed that, for $\beta_3(\infty) < \sqrt{\epsilon_1}k$ (Kretschmann geometry), the lower branch (Fig. 2) splits into two modes; the expected leaky Fano as well as another wave, which was bound above a cutoff film thickness h_{co} and was growing (with propagation distance) for $h_{co} > h$. Our results for this case show further that h_{co} $(\propto \epsilon_I^{-1})$ is independent of ϵ_3 (fixed ϵ_1) for $\sqrt{\epsilon_1}k > \beta_3(\infty)$ and that h_{co} (independent of ϵ_I) decreases with increasing ϵ_3 for $\beta_3(\infty) > \sqrt{\epsilon_1}k$ until $h_{co} = 0$ for $\epsilon_1 = \epsilon_3$. In the latter limit it corresponds to what is usually believed to be the bound, symmetric Fano mode; hence we conclude that the two solutions previously identified as Fano waves actually belong to different branches of the dispersion relation. In addition, we find that the leaky mode (s_l, Fig. 2) exists for all values of $\epsilon_1 \neq \epsilon_3$ (s_l in Fig. 3), even for $\beta_3(\infty) > \sqrt{\epsilon_1}k$. Therefore the lower branch splits into two distinct waves for all values of $\epsilon_1 \neq \epsilon_3$.

We also find that the upper branch (Figs. 2 and 3) consists of two modes, a leaky one (a_l) as well as the

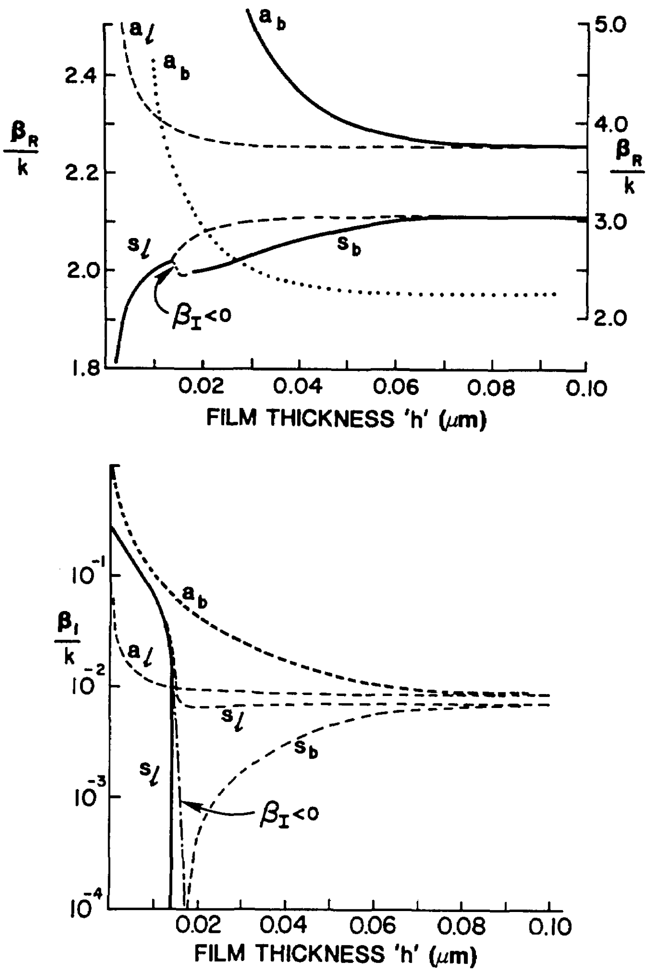

Fig. 3. The dispersion curves for a metal film of thickness h with $\epsilon_1 = 4.0$ and $\epsilon_3 = 3.61$, $\lambda = 0.633$ μm, and $\epsilon = -19 - 0.53i$; s and a refer to symmetric and antisymmetric modes, respectively. The dashed lines show the new features that appear when $\epsilon_I \neq 0$. The dotted line corresponds to the upper ($\epsilon_I = 0$) antisymmetric mode on a different scale. The subscripts b and l refer to bound and leaky modes, respectively.

usual bound Fano wave (a_b). It persists with $\beta_R > \beta_1(\infty)$ as long as $\epsilon_1 \neq \epsilon_3$. We note that each pair of modes becomes degenerate at $h \rightarrow \infty$, as required for uncoupled surface plasmons guided by widely separated surfaces.

The new leaky waves (a_l) that occur for $\beta_R > \beta_1(\infty)$ have a simple interpretation. They are mathematically characterized by fields that grow exponentially into the dielectric into which energy is radiated. They have meaning only in a limited region above the film and require some transverse plane (say, $x = 0$) containing an effective source that launches a localized wave at the ϵ_1–ϵ_m interface whose field decays exponentially across the film. For any plane $x > 0$, the wave radiates at some angle θ into medium ϵ_3, and the ϵ_3 field amplitude grows exponentially for only a finite distance z, which is a function of x and θ. Hence this wave is essentially the surface plasmon associated with a ϵ_1–ϵ_m interface bounded by semi-∞ media, which loses energy through radiation into medium ϵ_3. Assuming that the field at the ϵ_3–ϵ_m interface acts as an effective source, the diffraction integral predicts an angular spectrum $f(\theta) \propto [(\beta_R - \sqrt{\epsilon_3}k \sin\theta)^2 + \beta_I{}^2]^{-1}$. Since $\beta_R > \sqrt{\epsilon_3}k$, the radiation field has a broad angular spectrum. We note that dissipation in the metal is the dominant loss mechanism, and, because the leaky wave carries less energy in the metal than the bound antisymmetric mode, the attenuation of the leaky waves is less than for the bound mode (Figs. 2 and 3).

The leaky mode on the lower branch is a surface-plasmon wave associated with the ϵ_3–ϵ_m boundary between semi-infinite media that for finite films radiates into medium ϵ_1. For $\beta_R > \sqrt{\epsilon_1}k$, the angular spectrum of the radiation field is broad, and for $\beta_R < \sqrt{\epsilon_1}k$ it is sharply peaked at $\sin\theta = \beta_R/\sqrt{\epsilon_1}k$.

These new ($\beta_R > \sqrt{\epsilon_1}k, \sqrt{\epsilon_3}k$) leaky modes cannot be easily excited by means of prism or grating couplers that utilize a single input plane wave. Since the outgoing fields have a wide angular spectrum, efficient excitation requires a properly phased angular distribution of incident fields. They can, however, be excited by illuminating a transverse face of the sample with incident fields that match the transverse field distribution of semi-infinite media surface plasmons.[10] This approach is called end-fire coupling and is well known in integrated optics.

In summary, we have solved for the characteristic waves of a lossy, polariton-active film bounded by lossless dielectric media. New waves, both bound and leaky, have been identified in regions of the dispersion curves nominally associated with freely propagating radiation fields and bound modes, respectively. These new polariton waves can exist on a dielectric–metal film boundary because such a damped polariton is characterized by a finite wave-vector spectrum, which permits weak coupling to the continuum in the opposite dielectric medium. Although these effects were found numerically for modes guided by a metal film, the results should be valid for other cases in which surface-polariton propagation is a consequence of a negative dielectric constant.

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